

Coordinate-Free Assembly in Finite Element Methods: Application to 3D/2D/1D Coupling

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1 Introduction

The entirety of stiffness matrices of finite element method are computed evaluating a bilinear form associated to a differential problem for each coupled of basis functions.

Introduce the motivation for a coordinate-free approach: generality, elegance, abstraction, and applications in geometric numerical methods.

Ed è qui il punto chiave: la versione geometrica con cross: è più veloce; è più stabile; si vettorizza meglio; è più “fisica”. la versione M eye(4): è generale; è chiara; ma non si vettorizza in MATLAB. Se vuoi vettorizzare, hai fatto benissimo a tornare alla formula geometrica.

La tua presentazione è particolarmente adatta se:

lavori con inclusioni di dimensione diversa (1–2–3), vuoi una formulazione intrinseca (non legata allo spazio di embedding), vuoi enfatizzare l’assembly coordinate-free, vuoi giustificare un’implementazione generica.

Non enfatizzare troppo la “dimension-independent implementation”: presentala come una conseguenza naturale della formulazione geometrica. Se vuoi essere prudente, puoi aggiungere una frase tipo: For efficiency reasons, dimension-specific kernels may be used in practice, while the formulation remains dimension-independent.

Is the manuscript solid? Justification and structure

This section summarizes the rationale behind the manuscript, explains why the work is solid from a scientific perspective, and clarifies how the remaining parts (introduction, numerical results, and conclusions) can be naturally motivated.

Overall solidity of the paper

The manuscript is structurally solid for the following reasons.

- **Clear methodological contribution.** The work does not merely re-derive standard P1 finite element formulas. It introduces a geometric, coordinate-free formulation of local basis functions, their gradients, and the associated finite element assembly, valid in dimensions $d = 1, 2, 3$ and naturally extensible to lower-dimensional inclusions.
- **Mathematical consistency and necessity.** All mathematical tools introduced (Euclidean geometry, Gram matrices, dual bases, geometric definitions of gradients) are explicitly used in the subsequent derivations. There are no decorative definitions or unused abstractions.
- **Internal consistency checks.** Classical results are recovered as special cases (e.g. the Laplace operator in $d = 1$ and $d = 2$, cotangent-type formulas), providing strong validation of the framework.
- **Well-defined target audience.** The paper addresses researchers working on finite element methods for complex geometries, manifolds, and problems with lower-dimensional inclusions (e.g. fractures). For this audience, the motivation and relevance are immediately clear.

How to justify the work in the introduction

The introduction does not need to prove results; it should clearly answer three questions.

1. **What is the practical problem?** Finite element assembly on manifolds and in the presence of lower-dimensional inclusions is ubiquitous in applications such as fractured media, surface PDEs, and coupled multi-physics problems. However, implementations are often dimension-specific, coordinate-dependent, and difficult to generalize.
2. **Why is the current state of the art insufficient?** Many existing formulations rely on ad hoc expressions specialized to $d = 2$ or $d = 3$, making extensions to mixed-dimensional problems cumbersome and leading to code duplication and limited extensibility.
3. **What is the contribution of this work?** The manuscript proposes:
 - a unified geometric formulation valid in dimensions $d = 1, 2, 3$;
 - explicit expressions for basis functions and gradients derived directly from the element geometry;
 - a natural framework for finite element assembly in the presence of lower-dimensional inclusions.

Role of the numerical results

The purpose of the numerical section is not to provide large-scale benchmarks, but to demonstrate the correctness and applicability of the proposed framework.

A minimal and sufficient numerical validation includes:

- a standard diffusion (Laplace) problem in $d = 2$ or $d = 3$, verifying the expected convergence rate for linear finite elements;
- an example involving lower-dimensional inclusions (e.g. curves in 2D or fractures in 3D), illustrating the mixed-dimensional assembly procedure.

Optionally, one may include a qualitative comparison with dimension-specific implementations (e.g. in terms of code structure or extensibility).

Conclusions

The conclusions should briefly state that:

- a geometric and coordinate-free framework for finite element assembly has been presented;
- the approach applies uniformly to dimensions $d = 1, 2, 3$ and to problems with lower-dimensional inclusions;
- the formulation cleanly separates geometric information from algebraic assembly;
- the framework is well suited for extensions to higher-order elements, additional operators, and coupled multi-physics problems.

Final remark

This is not a fragile manuscript whose validity depends on benevolent referees. While editorial considerations (scope, length, journal focus) may affect acceptance, the scientific content is coherent, justified, and internally consistent. If not accepted in one venue, the work remains strong and defensible in others.

Remark on dimensional dependence. The explicit geometric constructions of the local basis functions and their gradients are necessarily dimension-dependent, as they rely on geometric operators that are specific to the intrinsic dimension of the element (e.g. length in $d = 1$, area and cross products in $d = 2$, oriented volumes and face normals in $d = 3$). This dependence is unavoidable and reflects the intrinsic nature of Euclidean geometry.

However, this dimensional dependence is strictly confined to the local geometric construction on a single element. Once the gradients of the basis functions are expressed in

geometric form, the subsequent formulation of the bilinear forms, the local stiffness matrices, and the global assembly procedure are uniform and independent of the intrinsic dimension.

In particular, the same abstract assembly framework applies to elements of different dimensions and to problems involving lower-dimensional inclusions, with the geometry entering exclusively through a finite set of dimension-specific local operators.

2 Euclidean geometry

2.1 Notation

Let $\mathbb{E}$ denote the three-dimensional Euclidean (affine) space and $\mathbb{V}$ the associated Euclidean vector space. Elements of $\mathbb{E}$, i.e., points, are denoted in sans-serif font, such as $\mathbf{P}$. Elements of $\mathbb{V}$, i.e., vectors, are denoted by letters with an overhead arrow, such as $\vec{v}$.

The operation $+$: $\mathbb{E} \times \mathbb{V} \rightarrow \mathbb{E}$ denotes the addition of a vector to a point. Given a point $\mathbf{X} \in \mathbb{E}$ and a vector $\vec{v} \in \mathbb{V}$, the expression

$$\mathbf{Y} = \mathbf{X} + \vec{v}$$

denotes the point $\mathbf{Y} \in \mathbb{E}$ obtained by adding the vector $\vec{v}$ to $\mathbf{X}$.

Similarly, the operation $-$: $\mathbb{E} \times \mathbb{E} \rightarrow \mathbb{V}$ defines the difference between two points. Given two points $\mathbf{X}, \mathbf{Y} \in \mathbb{E}$, the expression

$$\vec{v} = \mathbf{Y} - \mathbf{X}$$

yields the vector $\vec{v} \in \mathbb{V}$ directed from $\mathbf{X}$ to $\mathbf{Y}$.

Given $\vec{u}, \vec{v} \in \mathbb{V}$, we denote by $(\vec{u}, \vec{v})$ the scalar product in $\mathbb{V}$, by $\|\vec{v}\|$ the corresponding induced norm, and by $\vec{u} \wedge \vec{v}$ the cross product. Recall that the cross product is antisymmetric and satisfies

$$\vec{v} \wedge (\alpha \vec{v}) = \vec{0}, \quad \forall \alpha \in \mathbb{R}.$$

Therefore,

$$\vec{v} \wedge (\vec{u} + \alpha \vec{v}) = \vec{v} \wedge \vec{u}. \tag{1}$$

We also define the determinant operator

$$\det : \mathbb{V} \times \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{R},$$

which, for any three vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{V}$, is given by

$$\det(\vec{u}, \vec{v}, \vec{w}) = (\vec{u}, \vec{v} \wedge \vec{w}).$$

Properties of $\det$

- **Multilinearity.** The determinant is multilinear, i.e.,

$$\alpha \det(\vec{u}, \vec{v}, \vec{w}) = \det(\alpha \vec{u}, \vec{v}, \vec{w}) = \det(\vec{u}, \alpha \vec{v}, \vec{w}) = \det(\vec{u}, \vec{v}, \alpha \vec{w}),$$

and, in particular, it changes sign whenever one of its vector arguments is reversed.

- **Anticommutativity.** The determinant is anticommutative, i.e.,

$$\det(\vec{u}, \vec{v}, \vec{w}) = -\det(\vec{v}, \vec{u}, \vec{w}) = -\det(\vec{u}, \vec{w}, \vec{v}) = -\det(\vec{w}, \vec{v}, \vec{u}),$$

that is, it changes sign whenever two vector arguments are exchanged.

- **Cyclic invariance.** The determinant is unchanged under a circular shift of its arguments, i.e.,

$$\det(\vec{u}, \vec{v}, \vec{w}) = \det(\vec{v}, \vec{w}, \vec{u}) = \det(\vec{w}, \vec{u}, \vec{v}). \quad (2)$$

2.2 Length, area, and volume as geometrical operators

Given two points P_1 and P_2 in $\mathbb{E}$ defining a segment of length h , we have

$$\|P_1 - P_2\| = \|P_2 - P_1\| = h.$$

Given three non-collinear points P_1 , P_2 , and P_3 in $\mathbb{E}$, defining a triangle of area A , the vector

$$\vec{w} = (P_1 - P_2) \wedge (P_3 - P_2) \quad (3)$$

is orthogonal to the plane containing the triangle, and satisfies

$$\|\vec{w}\| = \|(P_1 - P_2) \wedge (P_3 - P_2)\| = 2A.$$

By cyclic permutation of the point indices, we also have

$$\vec{w} = (P_2 - P_3) \wedge (P_1 - P_3) = (P_3 - P_1) \wedge (P_2 - P_1). \quad (4)$$

Moreover, we have

$$\frac{1}{2A} \det(P_1 - P_2, P_3 - P_2, \vec{w}) = 2A.$$

In fact, due to the cyclic invariance (2), we obtain from the formula above

$$\frac{1}{2A} \det(\vec{w}, P_1 - P_2, P_3 - P_2) = \frac{1}{2A} (\vec{w}, \vec{w}) = 2A.$$

Finally, given four points P_1 , P_2 , P_3 , and P_4 in $\mathbb{E}$, no three of which are collinear, they define a tetrahedron of volume V . The determinant is related to the volume of the tetrahedron by

$$\det(P_1 - P_2, P_3 - P_2, P_4 - P_2) = \pm 6V. \quad (5)$$

Properties of the volume

- **Cyclic sign change.** A cyclic permutation of the point indices by one position $((1, 2, 3, 4) \mapsto (2, 3, 4, 1))$ produces a change of sign in the determinant, i.e.,

$$\det(\mathbf{P}_1 - \mathbf{P}_2, \mathbf{P}_3 - \mathbf{P}_2, \mathbf{P}_4 - \mathbf{P}_2) = -\det(\mathbf{P}_2 - \mathbf{P}_3, \mathbf{P}_4 - \mathbf{P}_3, \mathbf{P}_1 - \mathbf{P}_3).$$

- The vector

$$(\mathbf{P}_3 - \mathbf{P}_2) \wedge (\mathbf{P}_4 - \mathbf{P}_2)$$

is orthogonal to the plane containing the triangle with vertices $\mathbf{P}_2, \mathbf{P}_3, \mathbf{P}_4$. The determinant is positive if this vector points inward with respect to the tetrahedron, and negative if it points outward.

- Denoting by $\vec{w}_i$ the inward-pointing vector orthogonal to the face opposite $\mathbf{P}_i$, we have

$$(\mathbf{P}_i - \mathbf{P}_{i+1}, \vec{w}_i) = 6V, \quad \text{with cyclic indexing on } i.$$

Moreover, since $\mathbf{P}_{i+1}$ can be replaced by any vertex different from $\mathbf{P}_i$, we have, in general,

$$(\mathbf{P}_i - \mathbf{P}_j, \vec{w}_i) = 6V, \quad \text{for any } j \neq i.$$

3 Differential Geometry

Let $\mathbb{R}^d$ denote the d -dimensional coordinate space, where d is a positive integer. Elements of $\mathbb{R}^d$ are coordinate arrays and are denoted by bold letters, such as $\mathbf{x}$. The i -th entry of $\mathbf{x}$ is denoted by $x^i = (\mathbf{x})^i$, with $i = 1, \dots, d$.

Given a basis $\{\vec{e}_i\}_{i=1}^d$ of a d -dimensional subspace $\mathcal{W}^d \subseteq \mathbb{V}$, we define $\mathbf{v}^b \in \mathbb{R}^d$ by

$$(\mathbf{v}^b)^i = (\vec{v}, \vec{e}_i), \quad i = 1, \dots, d. \quad (6)$$

A basis also allows us to write any vector $\vec{v} \in \mathcal{W}^d$ as

$$\vec{v} = \sum_{j=1}^d v^j \vec{e}_j.$$

We furthermore introduce the array of vectors

$$\vec{E} = [\vec{e}_1 \ \dots \ \vec{e}_d],$$

that is, a d -dimensional array whose entries are the basis vectors. Interpreting $\vec{E}$ as a *row* array (notice the subscript indices) and $\mathbf{v}$ as a *column* array (notice the superscript indices), we may concisely write

$$\vec{v} = \vec{E} \mathbf{v}, \quad (7)$$

where the product above is intended as a row-by-column multiplication.

To determine the coordinates v^j , we take the inner product of (7) with $\vec{e}_i$, for $i = 1, \dots, d$. This yields the linear system

$$G \mathbf{v} = \mathbf{v}^b,$$

where G is the Gram matrix associated with the basis $\{\vec{e}_i\}$, defined by

$$G_{ij} = (\vec{e}_j, \vec{e}_i).$$

Thus, the basis introduces a bijection between $\mathcal{W}^d$ and $\mathbb{R}^d$, and the coordinates of $\vec{v}$ with respect to the basis $\{\vec{e}_i\}$ are given by

$$\mathbf{v} = G^{-1} \mathbf{v}^b.$$

Replacing this expression for $\mathbf{v}$ into (7), we obtain

$$\vec{v} = \vec{E}^b \mathbf{v}^b, \tag{8}$$

where

$$\vec{E}^b := \vec{E} G^{-1}$$

is an array of vectors $\vec{e}_i^b$. Again, the product is understood as a row-by-column multiplication, with $\vec{E}$ of size $1 \times d$ and G^{-1} of size $d \times d$.

For geometry enthusiasts, the vectors $\vec{e}_i^b$ (appearing as the elements of the array $\vec{E}^b$) are precisely the representatives in the space itself of the elements of the basis of the dual of $\mathbb{V}$ under the Riesz isomorphism. They are characterized by the property

$$(\vec{e}_i^b, \vec{e}_j) = \delta_{ij}. \tag{9}$$

3.1 Gradient and partial derivatives

Let $\Omega^d \subset \mathbb{E}$ be a smooth manifold of intrinsic dimension $d \leq 3$. In particular, Ω^1 denotes a curve, Ω^2 a surface, and Ω^3 a domain. Let $u : \Omega^d \rightarrow \mathbb{R}$ be a sufficiently smooth scalar function, and let $\mathcal{T}_{\mathbf{P}}^d \subseteq \mathbb{V}$ denote the tangent space at $\mathbf{P} \in \Omega^d$. We recall that $\mathcal{T}_{\mathbf{P}}^3 = \mathbb{V}$.

The gradient of u at a point $\mathbf{P} \in \Omega^d$, usually denoted by $\nabla u(\mathbf{P})$, is the unique vector $\vec{g} \in \mathcal{T}_{\mathbf{P}}^d$ such that

$$(\vec{g}, \vec{v}) = D_{\vec{v}} u(\mathbf{P}) \quad \forall \vec{v} \in \mathcal{T}_{\mathbf{P}}^d, \tag{10}$$

where $D_{\vec{v}} u(\mathbf{P})$ denotes the directional derivative of u along $\vec{v}$, given explicitly by

$$D_{\vec{v}} u(\mathbf{P}) = \lim_{\varepsilon \rightarrow 0} \frac{u(\mathbf{P} + \varepsilon \vec{v}) - u(\mathbf{P})}{\varepsilon}.$$

Remark 1. The gradient $\nabla u(\mathbf{P})$ is a vector in $\mathbb{V}$; its dimensionality does not depend on the intrinsic dimension d of the domain Ω^d .

Remark 2. If the directional derivative in the direction $\vec{v}$ can be written as

$$D_{\vec{v}}u(\mathbf{P}) = (\vec{g}, \vec{v}), \quad (11)$$

then the vector $\vec{g}$ is the gradient of u at $\mathbf{P}$.

Remark 3. If u is a linear function of the form

$$u(P) = (P - Q, \vec{g}), \quad (12)$$

for some fixed point $Q \in \mathbb{E}$, then $\vec{g}$ is the gradient of u .

Given now a basis $\{\vec{e}_i\}_{i=1}^d$ of $\mathcal{T}_{\mathbf{P}}^d$, the entries of $\mathbf{g}^b \in \mathbb{R}^d$ are

$$(\mathbf{g}^b)^i = (\vec{g}, \vec{e}_i) = D_{\vec{e}_i}u(\mathbf{P}), \quad i = 1, \dots, d.$$

To compute them explicitly, we introduce a coordinate system $\mathbf{x} = (x^1, \dots, x^d)$ centered at $\mathbf{P}$, such that a generic point $\mathbf{X} \in \Omega^d$ can be written as

$$\mathbf{X} = \mathbf{P} + \sum_{i=1}^d x^i \vec{e}_i, \quad (13)$$

so that $u(\mathbf{X}) = u(\mathbf{x})$. Then,

$$(\mathbf{g}^b)^i = D_{\vec{e}_i}u(\mathbf{P}) = \frac{\partial u}{\partial x^i}(\mathbf{P}). \quad (14)$$

4 Finite Element Assembly

In this section, we present the assembly of stiffness matrices associated with a family of widely used bilinear forms, including diffusion, advection, convection, and reaction operators, defined on the manifold Ω^d within a formal geometrical framework. Although the presentation is primarily focused on linear finite elements, which constitute the main objective of this manuscript, the proposed framework is fully general.

4.1 Bilinear forms

For a measurable subset Ξ , we denote by $H^1(\Xi)$ and $L^2(\Xi)$ the Sobolev space of order one and the Lebesgue space of square-integrable functions, respectively. Moreover, $C(\Xi)$ denotes the space of continuous functions on Ξ .

The bilinear forms of interest are defined as follows:

$$\begin{aligned}
a(u, v) &= \int_{\Omega^d} (\mathbf{K} \nabla u, \nabla v) \, d\Omega^d, \\
b(u, v) &= \int_{\Omega^d} (\vec{b}, \nabla u) v \, d\Omega^d, \\
c(u, v) &= \int_{\Omega^d} u (\vec{c}, \nabla v) \, d\Omega^d, \\
m(u, v) &= \int_{\Omega^d} \sigma u v \, d\Omega^d.
\end{aligned} \tag{15}$$

where u and v belong to a function space $V \subseteq H^1(\Omega^d)$. In weak formulations of differential problems, u represents the solution, while v is a test function.

The bilinear forms in (15) are associated with the diffusion, advection, convection, and reaction operators

$$-\nabla \cdot (\mathbf{K} \nabla u), \quad \vec{b} \cdot \nabla u, \quad -\nabla \cdot (\vec{c} u), \quad \sigma u,$$

respectively. The diffusion tensor $\mathbf{K}$, the advection field $\vec{b}$, the convection field $\vec{c}$, and the reaction coefficient σ are assumed to be sufficiently regular to ensure that the above bilinear forms are well defined on $H^1(\Omega^d)$. When the diffusion tensor coincides with the identity, the corresponding diffusion operator reduces to the Laplace operator $-\Delta u$, and the corresponding bilinear form is

$$a(u, v) = \int_{\Omega^d} (\nabla u, \nabla v) \, d\Omega^d.$$

4.2 Finite element discretization and assembly

We introduce a conforming mesh, or tessellation, $\mathcal{T}^d$, of the domain Ω^d , that is, a decomposition into $\mathcal{N}^d$ elements K , which are segments for $d = 1$, triangles for $d = 2$, and tetrahedra for $d = 3$. We further assume that the material properties $\mathbf{K}$, $\vec{b}$, $\vec{c}$, and σ , are constant on each element K .

The mesh is represented by a set of data structures consisting of a list of points, corresponding to the mesh vertices $\mathbf{P}_i$, together with a connectivity matrix $\mathcal{M}^d \in \mathbb{N}^{(d+1) \times \mathcal{N}^d}$. The ℓ -th column of $\mathcal{M}^d$, denoted by $\mathbf{m} = \mathcal{M}^d(:, \ell)$, contains the global indices of the vertices incident to the ℓ -th element $K \in \mathcal{T}^d$. In particular, m^i , with $i = 1, \dots, d+1$, denotes the global index associated with the local vertex index i of the element K .

The space of element-wise linear functions is defined as

$$X^d = \left\{ v \in C(\Omega^d) \mid v|_K \text{ is an affine function of } \mathbf{P}, \forall K \in \mathcal{T}^d \right\},$$

together with the (global) basis functions $\varphi_i \in X^d$, which are characterized by the interpolation conditions

$$\varphi_i(\mathbf{P}_j) = \delta_{ij},$$

where δ_{ij} denotes the Kronecker delta. For a given bilinear form $s(\cdot, \cdot)$, the associated matrix S is defined by

$$S_{ij} = s(\varphi_j, \varphi_i).$$

The assembly of the global stiffness matrix S associated with a bilinear form $s(\cdot, \cdot)$ is performed elementwise over the mesh by adding the contributions of the local stiffness matrices S^K , as summarized in Algorithm 1. The local stiffness matrix $S^K \in \mathbb{R}^{(d+1) \times (d+1)}$ is defined by

$$S_{ij}^K = s|_K(\varphi_{m^j}, \varphi_{m^i}), \quad i, j = 1, \dots, d+1,$$

where $s|_K$ denotes the restriction of the bilinear form s to the element K .

Algorithm 1 Assembly of the matrix S

```

1: Initialize  $S = 0$ 
2: for  $\ell = 1 : \mathcal{N}^d$  do                                 $\triangleright$  Loop over all elements of the tessellation  $\mathcal{T}^d$ 
3:    $\mathbf{m} = \mathcal{M}^d(:, \ell)$                                  $\triangleright$  Local-to-global map for the vertices of element  $K$ 
4:   for  $i = 1 : d+1$  do
5:     for  $j = 1 : d+1$  do
6:        $S_{ij}^K = s|_K(\phi_{m^j}, \phi_{m^i})$                      $\triangleright$  Assemble local stiffness entry
7:     end for
8:   end for
9:   for  $i = 1 : d+1$  do
10:    for  $j = 1 : d+1$  do
11:       $S_{m^i m^j} = S_{m^i m^j} + S_{ij}^K$                      $\triangleright$  Add local contribution to the global matrix
12:    end for
13:  end for
14: end for
```

Thus, introducing the local basis functions

$$\hat{\varphi}_i = \varphi_{m^i}|_K,$$

whose gradient satisfies

$$\nabla \hat{\varphi}_i(\mathbf{X}) = \nabla \varphi_{m^i}(\mathbf{X})|_K,$$

the entries of the local stiffness matrices associated with the bilinear forms of interest are

$$\begin{aligned}
A_{ij}^K &= \int_K (\mathbf{K} \nabla \hat{\varphi}_j(\mathbf{X}), \nabla \hat{\varphi}_i(\mathbf{X})) \, d\Omega^d, \\
B_{ij}^K &= \int_K (\vec{b}, \nabla \hat{\varphi}_j(\mathbf{X})) \hat{\varphi}_i(\mathbf{X}) \, d\Omega^d, \\
C_{ij}^K &= \int_K \hat{\varphi}_j(\mathbf{X}) (\vec{c}, \nabla \hat{\varphi}_i(\mathbf{X})) \, d\Omega^d, \\
M_{ij}^K &= \int_K \sigma \hat{\varphi}_j(\mathbf{X}) \hat{\varphi}_i(\mathbf{X}) \, d\Omega^d.
\end{aligned} \tag{16}$$

4.3 Assembly of the local stiffness matrices

The computation of the local stiffness matrices is carried out by introducing a local coordinate system on each element K , as described in (13). Let the vertices of K be denoted by $\mathbf{P}_1, \dots, \mathbf{P}_{d+1}$. The point $\mathbf{P}_1$ is chosen as the origin of the local coordinate system, and the basis vectors are defined as

$$\vec{e}_i = \mathbf{P}_{i+1} - \mathbf{P}_1, \quad i = 1, \dots, d.$$

Any point $\mathbf{X} \in K$ can then be represented in local coordinates $\mathbf{x} \in \mathbb{R}^d$ through

$$\mathbf{X} = \mathbf{P}_1 + \vec{E} \mathbf{x}, \quad x^i \geq 0, \quad \sum_{i=1}^d x^i \leq 1. \tag{17}$$

Accordingly, the local basis functions can be written as

$$\hat{\varphi}_i(\mathbf{x}) = \hat{\varphi}_i(\mathbf{X}), \quad i = 1, \dots, d+1.$$

Once the local coordinate system is introduced, the local basis functions are independent of the specific element K . For linear finite elements, the local basis functions admit the explicit expressions given below.

Case $d = 1$:

$$\hat{\varphi}_1(x^1) = 1 - x^1, \quad \hat{\varphi}_2(x^1) = x^1.$$

Case $d = 2$:

$$\hat{\varphi}_1(\mathbf{x}) = 1 - x^1 - x^2, \quad \hat{\varphi}_2(\mathbf{x}) = x^1, \quad \hat{\varphi}_3(\mathbf{x}) = x^2.$$

Case $d = 3$:

$$\hat{\varphi}_1(\mathbf{x}) = 1 - x^1 - x^2 - x^3, \quad \hat{\varphi}_2(\mathbf{x}) = x^1, \quad \hat{\varphi}_3(\mathbf{x}) = x^2, \quad \hat{\varphi}_4(\mathbf{x}) = x^3.$$

Denoting the gradient of $\hat{\varphi}_i$ by $\vec{g}_i$, from (8) it follows that

$$\vec{g}_i = \nabla \hat{\varphi}_i = \vec{E}^b \mathbf{g}_i^b,$$

where, from (14),

$$(\mathbf{g}_i^b)^j = \frac{\partial \hat{\varphi}_i}{\partial x^j}, \quad j = 1, \dots, d.$$

The explicit expressions of $\mathbf{g}_i^b$ in case of linear finite element basis functions are given below.

Case $d = 1$:

$$\mathbf{g}_1^b = [-1], \quad \mathbf{g}_2^b = [1].$$

Case $d = 2$:

$$\mathbf{g}_1^b = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \mathbf{g}_2^b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{g}_3^b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Case $d = 3$:

$$\mathbf{g}_1^b = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix}, \quad \mathbf{g}_2^b = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{g}_3^b = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{g}_4^b = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Finally, denoting by $\hat{K}$ the parametrization of the element K , the entries of the local stiffness matrices (16) can be explicitly computed via the following integrals

$$\begin{aligned} A_{ij}^K &= |K| \int_{\hat{K}} (\mathbf{K} \vec{E}^b \mathbf{g}_j^b, \vec{E}^b \mathbf{g}_i^b) d\mathbf{x}, \\ B_{ij}^K &= |K| \int_{\hat{K}} (\vec{b}, \vec{E}^b \mathbf{g}_j^b) \hat{\varphi}_i(\mathbf{x}) d\mathbf{x}, \\ C_{ij}^K &= |K| \int_{\hat{K}} \hat{\varphi}_j(\mathbf{x}) (\vec{c}, \vec{E}^b \mathbf{g}_i^b) d\mathbf{x}, \\ M_{ij}^K &= |K| \int_{\hat{K}} \sigma \hat{\varphi}_j(\mathbf{x}) \hat{\varphi}_i(\mathbf{x}) d\mathbf{x}. \end{aligned} \tag{18}$$

the integrals above, the parametrized local basis functions $\hat{\varphi}_i(\mathbf{x})$, and hence the vectors $\mathbf{g}_i^b$ containing their partial derivatives, as well as the reference element $\hat{K}$, do not depend on the specific element. Therefore, these quantities can be computed once for all elements. The dependence on the geometry of the element enters exclusively through the array $\vec{E}^b$, which depends only on the element K but not on the basis function.

5 Geometric assembly of local stiffness matrices

This section presents the two main results of this manuscript. First, in the case of linear finite elements, explicit expressions for the local basis functions and their gradients are obtained in a purely geometrical manner, relying only on the vertices of the element. As a consequence, the formulae (16) can be evaluated directly, without introducing the explicit parametrization (17).

Second, building on this observation, the vector entries of the array $\vec{E}^b$ are derived explicitly. This result is fundamental not only for linear finite elements, but for elements of arbitrary polynomial degree.

5.1 Geometric basis functions and gradients

In the following, the element K is either a segment, a triangle, or a tetrahedron. Its vertices are denoted by $P_1, \dots, P_{d+1}$. For convenience, we adopt a circular indexing convention, namely

$$P_{d+2} = P_1, \quad P_{d+3} = P_2, \quad \text{etc.}$$

Different equivalent formulations exist for the local basis functions on the element K . Although they coincide on K , their extensions to the full domain may differ. For completeness, several alternative expressions are collected in the Appendix.

5.1.1 Basis functions on a segment

Consider a segment element K defined by its two endpoints P_1 and P_2 , and let its length be $h = |K|$. We introduce the vectors

$$\vec{v}_i = P_i - P_{i-1}, \quad i = 1, 2,$$

where the circular convention implies $P_0 = P_2$. It follows immediately that $\vec{v}_1 = -\vec{v}_2$ and

$$h = \|\vec{v}_1\| = \|\vec{v}_2\|.$$

The local basis functions $\hat{\varphi}_i$, associated with the vertex P_i , is defined as

$$\hat{\varphi}_i(P) = \frac{(P - P_{i-1}, \vec{v}_i)}{(\vec{v}_i, \vec{v}_i)} = \frac{(P - P_{i-1}, \vec{v}_i)}{h^2}, \quad i = 1, 2.$$

These basis functions:

- are defined over the entire Euclidean space $\mathbb{E}$;
- are linear on K ;

- satisfy the Lagrangian condition

$$\hat{\varphi}_i(\mathbf{P}_j) = \delta_{ij}.$$

Since $\hat{\varphi}_i$ is linear, from Remark 3 its gradient is:

$$\vec{g}_i = \nabla \hat{\varphi}_i(\mathbf{P}) = \frac{\vec{v}_i}{h^2}, \quad i = 1, 2.$$

5.1.2 Basis functions on a triangle

Consider a triangular element K defined by its vertices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$, and let its area be $A = |K|$. Introduce the vectors

$$\vec{v}_i = \mathbf{P}_{i+2} - \mathbf{P}_{i+1}, \quad i = 1, 2, 3,$$

so that we have that the normal vector $\vec{w}$, as defined in (3), can be equivalently written (see (4)) as one of following cross product

$$\vec{w} = -\vec{v}_i \wedge \vec{v}_{i+1}, \quad i = 1, 2, 3.$$

The local basis functions $\hat{\varphi}_i$, associated with the vertex $\mathbf{P}_i$, is defined as

$$\hat{\varphi}_i(\mathbf{P}) = \frac{(\mathbf{P} - \mathbf{P}_{i+1}, \vec{v}_i \wedge \vec{w})}{(-\vec{v}_{i-1}, \vec{v}_i \wedge \vec{w})} = \frac{\det(\mathbf{P} - \mathbf{P}_{i+1}, \vec{v}_i, \vec{w})}{4A^2}, \quad i = 1, 2, 3.$$

These basis functions:

- are defined over the entire Euclidean space $\mathbb{E}$;
- are linear on K ;
- satisfy the Lagrangian condition

$$\hat{\varphi}_i(\mathbf{P}_j) = \delta_{ij}.$$

Since $\hat{\varphi}_i$ is linear, from Remark 3 its gradient is:

$$\vec{g}_i = \nabla \hat{\varphi}_i(\mathbf{P}) = \frac{\vec{v}_i \wedge \vec{w}}{4A^2}, \quad i = 1, 2, 3.$$

5.1.3 Basis functions on a tetrahedron

Consider a tetrahedral element K with vertices P_1, P_2, P_3, P_4 and volume $V = |K|$. Introduce the vectors

$$\vec{v}_i = P_{i+1} - P_i, \quad i = 1, 2, 3, 4.$$

The local basis functions $\hat{\varphi}_i$, associated with the vertex P_i , is defined as

$$\hat{\varphi}_i(P) = \frac{(P - P_{i+1}, \vec{v}_{i+1} \wedge \vec{v}_{i+2})}{(-\vec{v}_i, \vec{v}_{i+1} \wedge \vec{v}_{i+2})} = \frac{\det(P - P_{i+1}, \vec{v}_{i+1}, \vec{v}_{i+2})}{-\det(\vec{v}_i, \vec{v}_{i+1}, \vec{v}_{i+2})}, \quad i = 1, 2, 3, 4.$$

Because the denominator changes sign cyclically, it is convenient to define the vectors

$$\vec{w}_i = (-1)^i \frac{\vec{v}_{i+1} \wedge \vec{v}_{i+2}}{6V},$$

so that the local basis functions admit the compact form

$$\hat{\varphi}_i(P) = \frac{(P - P_{i+1}, \vec{w}_i)}{6V} \quad i = 1, 2, 3, 4.$$

These basis functions:

- are defined over the entire Euclidean space $\mathbb{E}$;
- are linear on K ;
- satisfy the Lagrangian condition

$$\hat{\varphi}_i(P_j) = \delta_{ij}.$$

Since $\hat{\varphi}_i$ is linear, from Remark 3 its gradient is:

$$\vec{g}_i = \nabla \hat{\varphi}_i(P) = (-1)^i \frac{\vec{v}_{i+1} \wedge \vec{v}_{i+2}}{6V} = \frac{\vec{w}_i}{6V}, \quad i = 1, 2, 3, 4.$$

5.2 Explicit expression for $\vec{E}^b$

We recall that, for any intrinsic dimension d of the manifold Ω^d , the gradients of the local basis functions satisfy

$$\vec{g}_i = \vec{E}^b \mathbf{g}_i^b, \quad i = 1, \dots, d+1, \quad (19)$$

where the arrays $\mathbf{g}_i^b \in \mathbb{R}^d$ are explicitly known:

- for $i = 2, \dots, d+1$, the vector $\mathbf{g}_i^b$ has all zero entries except a single 1 in position $i-1$;

- the remaining vector is

$$\mathbf{g}_1^b = \begin{bmatrix} -1 \\ \vdots \\ -1 \end{bmatrix} = - \sum_{j=2}^{d+1} \mathbf{g}_j^b.$$

From the identity (19), and from the structure of the arrays $\mathbf{g}_i^b$, we immediately obtain

$$\vec{g}_{i+1} = \vec{E}^b \mathbf{g}_{i+1}^b = \vec{e}_i^b, \quad i = 1, \dots, d,$$

where $\vec{e}_i^b$ denotes the i -th element of the array $\vec{E}^b$. Hence,

$$\vec{e}_i^b = \vec{g}_{i+1}, \quad i = 1, \dots, d.$$

Finally, we observe that these vectors satisfy the mutual orthogonality condition (9), as required by the geometric construction.

5.3 Summary and local stiffness matrices

For linear finite elements, the local basis functions admit explicit geometrical expressions depending only on the vertices of the element, as reported in Table 1. They can be written as linear functions of the physical point $\mathbf{P}$ and depend on a set of vectors $\vec{v}_i$ characterizing the geometry of the element. For $d = 2$, the construction involves a vector $\vec{w}$ orthogonal to the element K , whereas for $d = 3$ it involves vectors $\vec{w}_i$ orthogonal to the faces opposite to the vertices $\mathbf{P}_i$ and pointing inward to the element.

d	$ K $	$\vec{v}_i$	$\vec{w}/\vec{w}_i$	$\hat{\varphi}_i(\mathbf{P})$	$\nabla \hat{\varphi}_i = \vec{g}_i$
1	h	$\mathbf{P}_i - \mathbf{P}_{i-1}$		$\frac{(\mathbf{P} - \mathbf{P}_{i-1}, \vec{v}_i)}{h^2}$	$\frac{\vec{v}_i}{h^2}$
2	A	$\mathbf{P}_{i+2} - \mathbf{P}_{i+1}$	$-\vec{v}_1 \wedge \vec{v}_2$	$\frac{\det(\mathbf{P} - \mathbf{P}_{i+1}, \vec{v}_i, \vec{w})}{4A^2}$	$\frac{\vec{v}_i \wedge \vec{w}}{4A^2}$
3	V	$\mathbf{P}_{i+1} - \mathbf{P}_i$	$(-1)^i (\vec{v}_{i+1} \wedge \vec{v}_{i+2})$	$\frac{(\mathbf{P} - \mathbf{P}_{i+1}, \vec{w}_i)}{6V}$	$\frac{\vec{w}_i}{6V}$

Table 1: Geometric expressions of local basis functions and their gradients for $d = 1, 2, 3$.

Given these basis functions and their gradients $\vec{g}_i$, which are constant over the element, and using the identity

$$\int_K \hat{\varphi}_i d\Omega^d = \frac{|K|}{d+1},$$

the entries of the local stiffness matrices can be written explicitly as

$$\begin{aligned}
A_{ij}^K &= |K| (\mathbf{K} \vec{g}_j, \vec{g}_i), \\
B_{ij}^K &= \frac{|K|}{d+1} (\vec{b}, \vec{g}_j), \\
C_{ij}^K &= \frac{|K|}{d+1} (\vec{c}, \vec{g}_i), \\
M_{ij}^K &= \begin{cases} \frac{2|K|\sigma}{(d+1)(d+2)}, & \text{if } i = j, \\ \frac{|K|\sigma}{(d+1)(d+2)}, & \text{if } i \neq j. \end{cases}
\end{aligned} \tag{20}$$

It is instructive to examine the explicit form of the local stiffness matrices in the case of the Laplace operator, i.e., when the diffusion tensor $\mathbf{K}$ coincides with the identity, for $d = 1$ and $d = 2$.

Case $d = 1$. For a one-dimensional element K of length $h = |K|$, the local stiffness matrix associated with the Laplace operator takes the well-known form

$$A^K = \frac{1}{h} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

This expression coincides with the standard one-dimensional finite element stiffness matrix; the only difference is that the element length is explicitly identified with the measure of the element, i.e., $h = |K|$.

Case $d = 2$. The entries of the local stiffness matrix are given by

$$S_{ij} = \frac{1}{16A^4} (\vec{v}_j \times \vec{w}, \vec{v}_i \times \vec{w}) A$$

where $A = |K|$ is the area of the element.

Using the property of the scalar product between vector products, the numerator can be rewritten as

$$(\vec{v}_j \times \vec{w}) \cdot (\vec{v}_i \times \vec{w}) = (\vec{v}_j \cdot \vec{v}_i)(\vec{w} \cdot \vec{w}) - (\vec{v}_j \cdot \vec{w})(\vec{v}_i \cdot \vec{w}).$$

Since

$$\vec{w} \cdot \vec{w} = \|\vec{w}\|^2 = 4A^2, \quad \vec{v}_j \cdot \vec{w} = 0,$$

the above identity simplifies to

$$(\vec{v}_j \times \vec{w}) \cdot (\vec{v}_i \times \vec{w}) = 4A^2 (\vec{v}_j \cdot \vec{v}_i).$$

Therefore, the local stiffness matrix entries reduce to

$$S_{ij} = \frac{1}{4A} (\vec{v}_j \cdot \vec{v}_i).$$

This expression is consistent with classical results reported in the literature, such as the well-known cotangent formula and related efficient assembly techniques. It is worth emphasizing, however, that such formulations are specific to the Laplace operator. In contrast, the geometrical framework developed in this work applies to general diffusion operators.

6 Problems with lower-dimensional inclusions

Assembly on manifolds plays a fundamental role in two main classes of applications:

- diffusion processes on surfaces and elasticity models for shells and cloth;
- finite element discretizations of problems involving lower-dimensional inclusions.

The latter class represents the main application addressed in this manuscript, particularly in geoscientific contexts, where lower-dimensional inclusions typically correspond to fractures and injection or extraction wells.

Let $k = 2, 3$ denote the intrinsic dimension of the differential problem, and let $\Omega^k \subset \mathbb{E}$ be the domain on which the problem is defined. For each intrinsic dimension $d < k$, we assume the presence of $\mathcal{F}^d$ smooth inclusions (i.e., lower-dimensional manifolds) $\Omega_j^d \subset \Omega^k$, with $j = 1, \dots, \mathcal{F}^d$. For notational convenience, we set $\mathcal{F}^k = 1$ and identify $\Omega_1^k \equiv \Omega^k$.

Inclusions of the same intrinsic dimension may intersect but do not overlap, whereas inclusions of lower intrinsic dimension may intersect with, or be embedded within, those of higher intrinsic dimension. Typical configurations include:

- $k = 3$: a three-dimensional domain containing surfaces and curves;
- $k = 2$: a two-dimensional domain containing curves.

Let

$$W = \{v \in H^1(\Omega^k) \mid v|_{\Omega_j^d} \in H^1(\Omega_j^d), \forall d = 1, \dots, k-1, \forall j = 1, \dots, \mathcal{F}^d\},$$

and let $V \subset W$ be a subspace incorporating the Dirichlet boundary conditions. We consider the following weak problem: find $u \in V$ such that

$$s(u, v) = F(v), \quad \forall v \in V, \tag{21}$$

where F is a continuous linear functional on V .

The bilinear form $s(\cdot, \cdot)$ is defined as the sum of contributions over all manifolds,

$$s(u, v) = \sum_{d=1}^k \sum_{j=1}^{\mathcal{F}^d} s_j^d(u, v),$$

where each bilinear form $s_j^d(\cdot, \cdot)$ is defined on Ω_j^d and represents a suitable combination of the bilinear forms associated with the differential operators introduced in the previous sections.

6.1 Finite element assembly

Let $\mathcal{T}_1^k$ be a tessellation of Ω_1^k , which we assume to be conforming with respect to all inclusions. For each manifold Ω_j^d , the tessellation $\mathcal{T}_1^k$ induces a lower-dimensional tessellation $\mathcal{T}_j^d$. The associated connectivity matrices are denoted by

$$\mathcal{M}_j^d \in \mathbb{N}^{(d+1) \times \mathcal{N}_j^d},$$

where $\mathcal{N}_j^d$ is the number of elements in the tessellation $\mathcal{T}_j^d$.

The global stiffness matrix S is assembled by iterating over all intrinsic dimensions and inclusions, assembling the contributions associated with each bilinear form s_j^d , and adding them to the global matrix according to the corresponding connectivity information.

More precisely, for each pair (d, j) , a local matrix S_j^d is assembled over the finite element space $X_j^d \subset H^1(\Omega_j^d)$ defined on the tessellation $\mathcal{T}_j^d$. The resulting matrix is then added to the global matrix S according to the connectivity matrix $\mathcal{M}_j^d$, as summarized in Algorithm 2.

Algorithm 2 Assembly of the global matrix S with lower-dimensional inclusions

```

1: for  $d = 1, \dots, k$  do
2:   for  $j = 1, \dots, \mathcal{F}^d$  do
3:     assemble the local matrix  $S_j^d$  on  $\mathcal{T}_j^d$ 
4:     add  $S_j^d$  to the global matrix  $S$  using the connectivity matrix  $\mathcal{M}_j^d$ 
5:   end for
6: end for
```

A Local basis functions

The geometrical constructions of the local basis functions presented in the main text are not unique. Equivalent formulations can be obtained by exploiting different geometrical characterizations. In this appendix, explicit constructions are recalled for one- and two-dimensional elements.

A.1 Segment elements

The formulation presented in the main manuscript can be interpreted as a particular instance of basis functions of the form

$$\hat{\varphi}_i(\mathbf{P}) = \frac{(\mathbf{P} - \mathbf{P}_{i-1}, \vec{q})}{(\vec{v}_i, \vec{q})},$$

where $\vec{q} \in \mathbb{V}$ is an arbitrary vector such that $(\vec{v}_i, \vec{q}) \neq 0$. These functions therefore depend on the choice of the vector $\vec{q}$. In the construction adopted in the main text, this freedom is removed by selecting $\vec{q} = \vec{v}_i$.

An alternative, and more classical, expression of the local basis functions, related to barycentric coordinates, is given by

$$\hat{\varphi}_i(\mathbf{P}) = \frac{\|\mathbf{P} - \mathbf{P}_{i-1}\|}{h}.$$

Their gradient is then

$$\nabla \hat{\varphi}_i(\mathbf{P}) = \frac{1}{h} \frac{\mathbf{P} - \mathbf{P}_{i-1}}{\|\mathbf{P} - \mathbf{P}_{i-1}\|}.$$

However, for a point $\mathbf{P} \in K$, we have

$$\nabla \hat{\varphi}_i|_K = \frac{\vec{v}_i}{h^2},$$

which coincides with the expression of the gradient obtained in the main text.

A.2 Triangular elements

The formulation presented in the main manuscript can be interpreted as a particular instance of basis functions of the form

$$\hat{\varphi}_i(\mathbf{P}) = \frac{\det(\mathbf{P} - \mathbf{P}_{i+1}, \vec{v}_i, \vec{q})}{\det(-\vec{v}_{i-1}, \vec{v}_i, \vec{q})},$$

where $\vec{q} \in \mathbb{V}$ is an arbitrary vector such that $(\vec{v}_i, \vec{q}) \neq 0$. These functions therefore depend on the choice of the vector $\vec{q}$. In the construction adopted in the main text, this freedom is removed by selecting $\vec{q} = \vec{w}$.

An alternative, and more classical, expression of the local basis functions, related to barycentric coordinates, is given by $\mathbf{P}_i$ can be written as

$$\hat{\varphi}_i(\mathbf{P}) = \frac{\|(\mathbf{P} - \mathbf{P}_{i+1}) \times \vec{v}_i\|}{2A}.$$

Their gradient is then

$$\nabla \hat{\varphi}_i(\mathbf{P}) = \frac{\vec{v}_i \times ((\mathbf{P} - \mathbf{P}_{i+1}) \times \vec{v}_i)}{2A \|(\mathbf{P} - \mathbf{P}_{i+1}) \times \vec{v}_i\|}.$$

However, for a point $P \in K$, we have

$$\nabla \hat{\varphi}_i|_K = \frac{\vec{v}_i \wedge \vec{w}}{2A},$$

which coincides with the expression of the gradient obtained in the main text.